Machine Learning in Employee Attrition

Elpida Siandri, Zoi Kasapi, Efthimia Mavridou, Eleni Vrochidou, George A. Papakostas
MLV Research Group, Department of Informatics
Democritus University of Thrace
Kavala, Greece
elpisian1@cs.duth.gr, zoikasa2@cs.duth.gr, emavridou@cs.duth.gr, evrochid@cs.duth.gr, gpapak@cs.duth.gr

2025 1st International Conference on Computational Intelligence Approaches and Applications (ICCIAA) | 979-8-3315-2365-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCIAA65327.2025.11013566

***Abstract*—Employee attrition poses a significant organizational challenge, disrupting operations, escalating costs, and diminishing institutional expertise. Recent advancements in machine learning (ML) provide innovative solutions for identifying potential attrition risks and implementing preemptive interventions. This paper examines advanced ML approaches, including Random Forests, Gradient Boosted Trees, and Long Short-Term Memory network, highlighting their ability of processing intricate datasets and temporal trends. Additionally, the study indicates the critical role of Explainable AI (XAI) in ensuring interpretability and fostering ethical, transparent decision-making. By integrating multimodal data sources and utilizing privacy-preserving methods, organizations can derive actionable insights while maintaining compliance with ethical and regulatory standards. Research findings can serve as a roadmap for leveraging ML to enhance workforce retention strategies, optimize HR processes, and foster sustainable organizational growth.**

Keywords— workforce attrition, machine learning, predictive analytics, Explainable AI, employee retention, HR optimization, ethical AI

I. Introduction

Employee attrition, namely turnover, remains a persistent and critical challenge for organizations globally. This voluntary and involuntary employee departure phenomenon disrupts business operations, leads to high financial and operational costs, and diminishes institutional knowledge. High turnover results in lower team morale, reduced employee engagement and a poor work culture, culminating in a cycle of productivity and attrition declines. Thus, it has become a strategic imperative across industry sectors to address these challenges [1], [2].

Recent studies emphasize the role of highly developed machine learning (ML) algorithms, such as XGBoost and Artificial Neural Networks (ANNs), in improving predictive accuracy for employee attrition. These models leverage diverse datasets, capturing a wide array of factors from job satisfaction and workload to external market conditions, offering human resources (HR) teams enhanced precision in targeting at-risk employees [3], [4]. A recent study by Al Akasheh et al. [5] (2024) highlights the growing role of ML, providing evidence of its success in accurately predicting attrition through the use of advanced methodologies like deep learning.

ML has revolutionized employee attrition prediction by uncovering complex patterns that traditional methods often miss. By identifying employees at risk of leaving, ML enables HR teams to implement proactive measures, such as personalized engagement programs, tailored professional development opportunities, and competitive compensation adjustments. This data-driven approach both enhances employee retention and fosters a more resilient and satisfying workforce [5], [6]. By integrating multimodal data sources and utilizing privacy-preserving methods, organizations can derive actionable insights while maintaining compliance with ethical and regulatory standards.

Based on the above, the prime objective of this paper is the evaluation of the effectiveness of advanced ML models in predicting employee turnover. Traditional statistical approaches, such as logistic regression, have served as foundational tools in attrition prediction; however, recent advancements in ML have significantly enhanced predictive accuracy and reliability. Ensemble methods, including Random Forests, Gradient Boosted Trees, and stacking approaches, integrate the strengths of multiple algorithms, resulting in superior performance. Neural networks (NNs) as Long Short-Term Memory (LSTM) models, further demonstrate their value in handling temporal employee data, providing actionable insights that support data-driven decision-making [1], [7]–[9]. Additionally, the integration of multimodal data—combining structured HR metrics with unstructured sources such as employee feedback and sentiment analysis—expands the predictive power of these models, enabling a more holistic understanding of attrition drivers [1], [10].

Sharma (2021) [11] emphasizes that using multimodal datasets, which integrate behavioral and qualitative data, leads to more accurate predictions and helps uncover hidden factors influencing employee turnover. Another critical focus of this study is the ethical and practical challenges associated with deploying ML systems in HR practices. Attrition datasets often suffer from data sparsity and imbalance, making it challenging to model outcomes like high-performer resignations accurately. Addressing these issues requires advanced preprocessing techniques, such as the Synthetic Minority Over-sampling Technique (SMOTE) towards class balancing and transfer learning to adapt pre-trained models to HR-specific contexts [10], [13]. Addressing the ethical worries regarding the utilization of sensitive employee data is another core objective. These methods allow organizations to adopt advanced ML solutions responsibly, ensuring alignment with both legal standards and company values [14], [15].. Privacy-preserving techniques, including federated learning and anonymization protocols, play a pivotal role in ensuring compliance with data protection regulations and safeguarding employee trust [6], [16], [17].

Emerging trends in ML-driven attrition analysis further enhance the applicability and relevance of these technologies. Explainable Artificial Intelligence (XAI) frameworks, like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), empower HR practitioners to interpret predictions with transparency, ensuring alignment with organizational values and ethical standards. These tools not only improve trust in ML models but also support accountability in HR decision-making [18], [19]. Moreover, innovations such as edge computing and lightweight ML models enable real-time, decentralized analytics, making attrition management more efficient and

Authorized licensed use limited to: University of Thrace (Democritus University of Thrace). Downloaded on November 07,2025 at 19:20:19 UTC from IEEE Xplore. Restrictions apply.
979-8-3315-2365-7/25/$31.00 ©2025 IEEE

adaptable to dynamic organizational environments [20], [21]. By addressing these objectives, this work aims to aid towards developing ethical, transparent, and effective ML-driven solutions for employee attrition. The findings of the current work aim to advance both theoretical and practical knowledge, equipping HR professionals with innovative tools to enhance employee retention strategies while upholding fairness, accountability, and data privacy standards.

The remaining of this work is organized as follows. Research methodology is presented in Section II. Fundamental concepts for the comprehension of the conducted study are provided in Section III, referring to relevant contributions of ML methods. Section IV reviews supervised, and unsupervised ML methods and deep learning methods for churn prediction. Current trends and challenges are highlighted in Section V, while the role of datasets is referenced in Section VI. Finally, Section VII presents future prospects, and VIII concludes the paper.

II. Research Methodology

This study adopts a systematic and structured methodology to investigate the application of ML techniques in predicting employee attrition. By synthesizing existing literature, it aims to derive high-quality insights into theoretical foundations and advanced methodologies that address employee turnover.

More specifically, the study adhered to the PRISMA guideline towards ensuring transparent and reproducible results. The process began with an extensive database search using the specified keywords. Titles and abstracts of the search results were screened to eliminate irrelevant or duplicate entries, followed by a detailed assessment of full-text articles based on the inclusion criteria. The selected studies were categorized into three domains: theoretical frameworks, applied research, and comparative analyses.

Theoretical frameworks explored foundational ML concepts, including supervised and unsupervised learning techniques. These studies provided insights into the mechanics of ML algorithms and their adaptability to HR contexts. Applied research demonstrated real-world implementations of ML models across industries such as retail, healthcare, and technology, showcasing practical approaches to attrition prediction. Comparative analyses evaluated the strengths and limitations of different ML techniques.

The primary sources of data for this review include Scopus and Google Scholar, selected for their extensive repositories of peer-reviewed academic works and high-impact research articles. The search utilized the query "employee attrition" AND "machine learning", designed to capture studies encompassing both foundational theories and practical applications. Scopus provided 417 articles, offering a foundation of theoretical and applied research. Google Scholar added 138 articles, providing less conventional but complementary resources to enhance the diversity of the dataset.

The research process was further supplemented by publicly available datasets, such as the IBM HR Analytics dataset, which offered empirical evidence to validate findings and benchmark ML models. These datasets were instrumental in enabling comparisons of predictive performance across studies [16]. Additionally, research conducted by Geiler et al. (2022) [22] suggests that the use of datasets like IBM HR Analytics improves the reliability of attrition prediction models by incorporating broader employee data.

To ensure methodological rigor, inclusion criteria were meticulously defined. The review focused on peer-reviewed articles and conference proceedings published between 2015 and 2024, capturing recent advancements in ML techniques. Priority was given to studies employing state-of-the-art methodologies, including deep learning, ensemble methodologies, and transformer-based architectures, which have demonstrated superior performance in predictive modeling [18], [23]. Particular attention was given to research utilizing hybrid ML models, such as CatBoost and Random Forests, which integrate traditional statistical methods with advanced algorithms. These hybrid approaches consistently outperformed standalone models, providing actionable insights into the multifaceted nature of employee attrition [3], [12].

Each study was evaluated against three criteria: methodological robustness, citation impact, and relevance. Robustness included an assessment of data preprocessing techniques, feature engineering, and validation methods. Studies using advanced cross-validation methods and benchmark datasets, such as the IBM HR Analytics dataset, were given higher consideration. Citation impact served as an indicator of research influence, while relevance ensured a focus on predictive modeling for employee attrition or related workforce management strategies [1], [24]. Moreover, Nawaz et al. (2024) [25] emphasize the importance of using benchmark datasets in predicting attrition effectively, as they allow for objective comparisons across studies and enhance the validity of findings. Therefore, to ensure depth and relevance, the 48 most impactful articles were selected from these sources based on methodological robustness, citation impact, and relevance to the study's objectives. The breakdown of articles per database is presented in Table I.

TABLE I. Selected Articles per Database

Database	Articles found	Articles selected	References
Scopus	417	41	[1]–[4], [6]–[10], [12]–[16], [18]–[21], [24], [26]–[47]
Google Scholar	138	7	[5], [11], [17], [22], [23], [25], [48]

III. Definitions and Frameworks and ML Contributions

The phenomenon of employee attrition covers all aspects of the exit of an employee and has a substantial effect on the morale of the team, the efficiency of operations, and the overall productivity of the organization. Recent studies emphasize that attrition rates vary significantly across industries, with sectors like technology and healthcare facing the highest turnover due to rapid innovation cycles and workforce demands. These industry-specific insights underline the importance of tailoring retention strategies to sectoral challenges [15], [26]. In the following, fundamentals such as types of attrition and factors that influence it, are briefly reviewed.

A. Types of Attrition

Attrition may be voluntary or involuntary. Voluntary attrition occurs when employees initiate their departure, often

Authorized licensed use limited to: University of Thrace (Democritus University of Thrace). Downloaded on November 07,2025 at 19:20:19 UTC from IEEE Xplore. Restrictions apply.

driven by dissatisfaction with their roles, inadequate compensation, limited career growth opportunities, or a mismatch with organizational culture. External factors, such as the availability of better opportunities or personal aspirations, also contribute. Research highlights that voluntary attrition often signals internal organizational issues, necessitating timely intervention from HR teams to address these challenges and improve workplace conditions [2], [16], [18], [44]. Studies leveraging ML techniques, such as sentiment analysis of employee feedback, have identified dissatisfaction with leadership and communication gaps as significant drivers of voluntary attrition. These insights allow organizations to focus on enhancing leadership training and internal communication frameworks [14], [48]. The work of Seixas et al. (2023) [34] further confirms the latter, noting that sentiment analysis can reveal deep insights into the underlying causes of employee dissatisfaction, helping organizations address issues before they escalate.

In contrast, involuntary attrition results from the employer's decision to terminate an employment relationship. This may occur due to layoffs driven by financial constraints, downsizing during organizational restructuring, or dismissals related to employee performance. While such decisions often align with the company's strategic goals, they can adversely affect the morale of the remaining employees and harm the organization's external reputation. Transparent communication and clearly articulated policies are critical for mitigating these negative effects and maintaining a cohesive workforce [6]. ML models have been employed to analyze involuntary attrition trends, highlighting patterns that indicate the need for more equitable performance evaluation systems. For example, bias detection algorithms in performance reviews have been instrumental in improving fairness and reducing unnecessary turnover [33], [40]. Nawaz et al. (2024) [25] emphasize the effectiveness of bias detection algorithms, which help to ensure fairer employee evaluations, ultimately reducing the likelihood of involuntary attrition.

B. *Influence Factors*

The factors leading to attrition are broadly assessed as external or internal. External factors relate to macroeconomic trends, industry dynamics, and labor market conditions. At times of an economic upswing, for example, employees are less afraid to venture into new opportunities and the attrition rate is higher. In contrast, during economic downturns attrition rates tend to fall based on the perceived lack of external options [1].

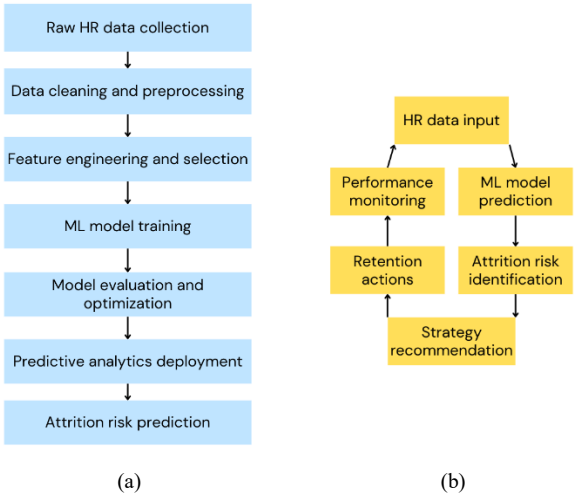
Advanced ML tools, such as predictive analytics platforms, have been used to assess the impact of external factors like regional economic growth rates and industry competitiveness. These tools enable organizations to align retention strategies with external market dynamics effectively [4], [32]. Research by Geiler et al. (2022) [22] further highlights the importance of integrating external market conditions into attrition models, allowing HR professionals to better anticipate external factors influencing workforce turnover.

Internal factors like compensation structures, workplace culture, and quality of leadership are specific to the organization itself. Low pay, lack of recognition, and poor relationships with co-workers are consistently among the best predictors of turnover. Lack of support systems, poor organization of work and organizational values can also trigger employee dissatisfaction. Focusing on internal access is the key to creating a culture that is supportive and happy [2], [16].

The integration of employee sentiment analysis, using tools such as NLP, has enabled HR teams to proactively identify dissatisfaction trends. For instance, analyzing open-ended responses in surveys provides deeper insights into employee concerns about management practices or workplace inclusivity. [39][43]. The study by Al Akasheh et al. (2024) [5] discusses how NLP tools can be utilized to parse employee feedback, providing HR teams with actionable data to address the underlying causes of dissatisfaction. Sharma (2021) [11] highlights that NLP tools, when applied to employee feedback, can help HR teams detect dissatisfaction and proactively address concerns before they escalate into attrition.

C. *Predictive Analytics*

It is essential to analyze attrition patterns to work towards data-driven HR strategies that can help address the root causes of employee turnover. ML based predictive analytics have emerged as a powerful game-changing tool in this space through the analysis of past and current employee data. For example, predictive models are able of identifying employees possible of leaving by relying on patterns like a drop in performance metrics, increased absenteeism, or feedback indicating dissatisfaction. These models empower HR teams to implement targeted interventions, such as enhanced career development programs, revised compensation packages, or improved employee engagement initiatives. Fig. 1 presents the analysis process in employee attrition prediction while Fig. 2 illustrates how ML models integrate into HR workflows. The pipeline begins with HR data input, followed by machine learning-based attrition prediction, risk identification, and strategic recommendations. The HR team then implements targeted retention actions and monitors performance, creating a continuous feedback loop to refine workforce management strategies.



(a) (b)

Fig. 1. (a) Data processing in employee attrition prediction; (b) HR model integration workflow.

Furthermore, identifying trends within specific departments or demographics enables organizations to address systemic issues proactively [1]. Latest developments in deep learning models, such as recurrent neural networks (RNNs), demonstrated superior performance in analyzing temporal data, enabling the identification of long-term attrition risks based on employee history. These techniques improve

Authorized licensed use limited to: University of Thrace (Democritus University of Thrace). Downloaded on November 07,2025 at 19:20:19 UTC from IEEE Xplore. Restrictions apply.

precision and offer actionable insights for HR teams [15], [26]. As Nawaz et al. (2024) [25] highlight, deep learning models such as RNNs excel at capturing long-term trends in employee behavior, which is crucial for predicting turnover based on evolving patterns in employee engagement and performance. Recent advancements in deep learning models, such as recurrent neural networks (RNNs), have shown superior performance in analyzing temporal data, enabling the identification of long-term attrition risks based on employee history. These techniques improve precision and offer actionable insights for HR teams [15], [26]. As Nawaz et al. (2024) [25] highlight, deep learning models such as RNNs excel at capturing long-term trends in employee behavior, which is crucial for predicting turnover based on evolving patterns in employee engagement and performance.

Emerging studies also highlight the value of advanced analytics, particularly XAI, in enhancing the interpretability of predictive models. XAI techniques such as and LIME provide HR professionals with insights into the factors influencing predictions, ensuring fairness, transparency, and accountability in decision-making processes. Privacy-preserving approaches, including federated learning and edge computing, further enhance the security and ethical alignment of predictive analytics by safeguarding sensitive employee data while maintaining analytical accuracy [37]. Seixas et al. (2023) [34] also stress the importance of XAI in improving transparency in ML-driven decision-making, ensuring that HR managers understand the rationale behind attrition predictions and can act accordingly.

D. Strategic Implications

By categorizing attrition into its voluntary and involuntary components and analyzing its external and internal drivers, organizations can gain a comprehensive understanding of workforce dynamics. This understanding forms the foundation for implementing proactive, data-driven HR practices that not only reduce attrition-related costs but also enhance employee satisfaction and retention. Leveraging predictive analytics and integrating multimodal data—combining structured HR metrics with unstructured sources such as employee feedback—allows organizations to address attrition challenges holistically and effectively. These insights empower HR teams to implement systemic improvements and maintain a competitive edge in the talent market [44], [45].

Incorporating ML-driven tools into strategic HR planning also allows organizations to simulate retention outcomes based on hypothetical interventions, such as altering compensation structures or introducing flexible work policies. These simulations provide invaluable foresight for decision-making [12], [48].

IV. ML IN CHURN PREDICTION

ML has become a valuable tool for the prediction of employees turnover, offering the ability to analyze large datasets and generate accurate predictions regarding employee behavior and potential attrition. Three main categories are identified, comprising (1) Supervised (2) unsupervised and (3) deep learning methods.

A. Supervised Learning

Supervised learning is considered as an effective approach for predicting employee churn. This approach relies on datasets with labeled outcomes, such as whether an employee has stayed or left the organization. By training models on data that includes both employee characteristics and corresponding churn labels, algorithms can learn to identify patterns and relationships that contribute to turnover. Two of the most widely used methods in this context are Random Forest and Neural Networks, both of which have demonstrated excellent performance in predicting employee retention outcomes.

Random Forest, an ensemble-based method, works by combining predictions from multiple decision trees to produce a final, aggregated result. This approach is particularly useful for substantial datasets, as it is both highly accurate and resistant to overfitting. Research highlights that Random Forest is adept at capturing nonlinear relationships within employee data, such as the connections between engagement levels, stress, and resignation risks. Its robust performance across diverse datasets makes it an invaluable tool for HR analytics [38].

Neural Networks, on the other hand, are particularly effective for analyzing complex and nonlinear relationships in data. These models utilize multiple layers of neurons to uncover intricate patterns that may not be immediately apparent. Neural Networks are especially valuable in cases where employee churn is influenced by multifaceted factors, such as professional growth opportunities, job satisfaction, and team dynamics. Studies emphasize that these models excel in high-complexity scenarios, often outperforming other methods in terms of predictive accuracy when hidden variables and subtle correlations play a significant role [21], [23].

Comparative analyses between Random Forest and Neural Networks highlight their unique strengths. While Random Forests are acknowledged for their speed and capability of handling big datasets efficiently, Neural Networks are better suited for scenarios requiring the identification of hidden parameters and complex linkages. This makes them an ideal choice for tasks involving intricate data structures and high-stakes predictions [8], [17]. In practice, combining these models has proven to be particularly effective. When used together, Random Forest and Neural Networks complement each other, addressing gaps in individual performance and improving overall predictive capabilities. This hybrid approach is especially useful in cases where datasets are incomplete, or when the features and parameters vary significantly. By leveraging the strengths of both methods, organizations can gain deeper insights into employee churn and devise more targeted retention strategies [17].

B. Unsupervised Learning

Unsupervised learning provides a compelling approach to discovering suppressed patterns, relationships, and structures within employee data, without requiring predefined labels. Unlike supervised methods, unsupervised techniques enable the exploration of underlying trends and connections in data. This allows for the identification of significant patterns that may not be immediately evident and offers a broader understanding of factors influencing employee turnover.

Advanced clustering methods, such as Density-Based Spatial Clustering (DBSCAN), are increasingly being adopted to identify outliers or unique employee behaviors that could signify hidden attrition risks. These techniques complement traditional clustering methods by focusing on density patterns rather than fixed groupings, offering a more dynamic analysis of workforce data [4], [32]. Geiler et al. (2022) [22] emphasize that advanced clustering methods such as DBSCAN are

Authorized licensed use limited to: University of Thrace (Democritus University of Thrace). Downloaded on November 07,2025 at 19:20:19 UTC from IEEE Xplore. Restrictions apply.

critical for understanding employee behavior, as they allow organizations to detect unusual patterns that may otherwise go unnoticed.

Clustering is a common application of unsupervised learning in this context. A well-known clustering algorithm is k-means, which can group data into clusters based on their similarity. By calculating the Euclidean distance between data points, k-Means organizes employees into groups that share common behavioral or risk characteristics. This method is particularly effective for large datasets, as it enables HR teams to identify employee segments at high risk of leaving. These clusters can reveal patterns related to churn, such as low engagement or dissatisfaction, that would be difficult to detect otherwise [17], [27].

Another important technique is hierarchical clustering, which creates a hierarchy of clusters using two distinct methods: agglomerative and divisive. The agglomerative approach begins with each employee as an individual cluster and merges them progressively based on similarities, while the divisive approach starts with all employees in a single cluster and divides them into smaller subgroups. This technique is particularly advantageous for understanding complex relationships between clusters, as it provides a detailed structure called a dendrogram. Contrasting k-means, hierarchical clustering methods do not demand a predefined number of clusters, making it well-suited for analyzing diverse and overlapping traits among employees. In HR analytics, this method helps identify subgroups of employees who share similar attributes linked to turnover risk, such as job satisfaction or stress levels [13]. The application of unsupervised learning methods to HR data offers a promising way to address employee turnover. While supervised approaches rely on labeled data, such labels are often unavailable in real-world scenarios. By uncovering the fundamental structures and patterns within the data, unsupervised techniques enable organizations to gain insights into employee groups that might otherwise remain hidden. These groups may include employees with low engagement, high stress, or signs of disillusionment.

C. Deep Learning

Deep learning models such as RNNs and LSTM are widely applied in data sources whose successive elements have a time-related or sequence-related connection to each other, allowing for pattern discovery and predicting the next state [10]. RNNs and LSTM are some of the most widely used deep learning architectures for sequential data modeling. Data, such as time series, in turn, can be modeled using RNN which is an architecture that processes sequential data, by keeping information learned from previous time steps and using that knowledge to make predictions in the future. While RNNs can learn from a sequence, struggle to memorize valuable things across long time periods as the gradients start vanishing after the computation, which is why LSTM is used, which is a special case of RNN [30].

RNNs and LSTMs struggle mostly with data complexity since the data usually has a lot of fluctuations and relationships between time points. However, they are still basic models thoroughly used in any situation such as stock price prediction, NLP, prediction problems or log data exploration where time series are treated since they form excellent functionalities for pattern identification and predictions.

V. Current Trends and Challenges

Regarding current trends, predictive analytics are increasingly used in this area to identify which employees are at risk of leaving and more and more AI tools find their place within classical HR management systems. However, these practices have allowed organizations to derive insights from workforce dynamics that, in turn, can inform actions to retain talent. Additionally, there has been a growing trend in the adoption of cloud-based AI platforms enabling enterprises efficiently processing and analyzing massive amounts of data while minimizing infrastructure costs. As Sharma (2021) [11] notes, AI platforms in HR are becoming essential for processing large datasets and making data-driven decisions that enhance employee retention. Another notable trend is the incorporation of edge computing, which allows the simultaneous processing of data directly at the source, reducing latency and enhancing the responsiveness of HR analytics systems. Additionally, scalability remains a challenge for XAI applications. Many HR systems process vast amounts of data daily, and integrating complex explainability tools can slow down performance. This is particularly useful in environments requiring immediate insights, such as high-turnover industries. Geiler et al. (2022) [22] also emphasize that edge computing enables HR specialists to conclude to faster, more data-based decisions for employees retention, especially in industries with high attrition rates.

A key challenge in applying these models is managing imbalanced datasets, which often result in biased predictions. Techniques like SMOTE have been employed to address this issue by balancing class distributions and enhancing model fairness. However, ensuring that these adjustments do not compromise data integrity remains a critical concern [26], [33]. Nawaz et al. (2024) [25] highlight the challenges of managing imbalanced datasets, noting that while techniques like SMOTE improve fairness, they can also introduce new challenges related to model generalization and accuracy.

Hence, the use of AI over contemporary business processes has a multitude of challenges. There are several hindering factors, one being the data bias. It is a well-known fact that data on which the models are trained may also introduce some background assumptions or biases, which can significantly alter how the algorithms work. For example, metrics used in historical data could recirculate outdated stereotypes at a consortium or institutional level; a specific example is employee churn models from where the outcomes can be biased. Therefore, it is important to find and minimize these biases and make the HR strategies more precise. As Seixas et al. (2023) [34] explain, mitigating data bias in AI models is crucial for ensuring fair and accurate predictions, particularly in sensitive areas like employee retention.

The other big hurdle is the interpretability of models. While RNNs and LSTMs are proven accurate predictive algorithms, they operate in black-box mode, meaning even if a person can name the model architecture, they have no more insight into their decision-making process. This is especially relevant in the domain of human resources where the interpretability of a model can lead to informing strategic initiatives like attrition prevention. The need arises for models that are not only accurate but also explainable and this has received greater emphasis with the rise of XAI techniques that can explain your decision-making process [16]. However, XAI techniques introduce computational complexity and

Authorized licensed use limited to: University of Thrace (Democritus University of Thrace). Downloaded on November 07,2025 at 19:20:19 UTC from IEEE Xplore. Restrictions apply.

require significant processing power and time, making real-time decision-making more difficult. Even when XAI tools provide interpretable insights, HR professionals may lack the technical expertise to understand and act on them effectively. Without proper training, HR teams might misinterpret model explanations or fail to integrate them into decision-making.

Moreover, ensuring data privacy and security remains a persistent challenge. Techniques to preserve privacy, like federated learning, have been developed to address these concerns by allowing decentralized data analysis without exposing sensitive employee information. However, their implementation requires robust computational resources and expertise, which may not be readily available in all organizations [12], [15]. Al Akasheh et al. (2024) [5] further explore the challenges of implementing privacy-preserving techniques in HR systems, highlighting that while federated learning offers significant benefits, it demands significant infrastructure and expertise to ensure secure and ethical data handling.

VI. The Role of Datasets

The dataset which is presented by Iparraguirre-Villanueva et al. [31], includes 29 features related to employee attrition, covering various aspects such as business travel, education, age, department, distances from home, and more. Each feature can significantly influence the decision of an employee to stay or leave, and these data were analyzed using ML models towards identifying the upmost critical factors for attrition. The inclusion of environmental and job satisfaction metrics in this dataset highlights the importance of qualitative attributes in attrition prediction. Studies have shown that qualitative factors, such as employee engagement and managerial support, are often the strongest predictors of turnover. This underscores the need for HR teams to go beyond demographic and performance data to understand employee sentiment [14]. As Seixas et al. (2023) [34] highlight, including qualitative data like employee engagement and managerial support is critical in predicting turnover, as these factors often outweigh traditional quantitative measures. The distribution of the data may not be evenly balanced, with more employees staying than leaving, which could impact model training. To address this, techniques for managing imbalances may be necessary [31]. The results of the employee attrition prediction models, illustrated in Fig.2, show strong performance, especially with Random Forest and XGBoost, which reported accuracies up to 98.7% and 98.8%, respectively. The latter implies their effectiveness to predict employee turnover. On the other hand, the NuSVC model, underperformed significantly, suggesting it may not be as reliable for this task. Overall, results indicate that the dataset is strong and well-suited for predicting employees attrition.

The IBM HR Analytics Employee Attrition Performance dataset [47] comprises 35 features, including both numerical and categorical features. These features span multiple facets of an employee's experience, including age, gender, job role, education, job satisfaction, work-Life balance, distance from home, salary, department, business travel frequency, overtime, marital status, and more. It employs ML methods like Logistic Regressors, Random Forests and Support Vector Machine to predict the probability of an employee leaving. This dataset also integrates advanced preprocessing steps, such as feature scaling and categorical encoding, which enhance the models' ability to generalize across diverse employee groups. These preprocessing steps are critical for models like SVM, which are sensitive to feature scaling [3], [41]. In this dataset, there is also an imbalance between employees who stay and leave. Out of the models, Random Forest was the most powerful one, achieving high accuracy while other models like Logistic Regression and SVM gave good results as well. The SVC and K-Neighbors Classifier reported optimal performance of 89.1%, with XGB Classifier close behind at 88.8%. Accuracy results for all used models are included in Fig. 3. These results suggest that while most models performed moderately to well, some were more effective than others for predicting employee attrition.

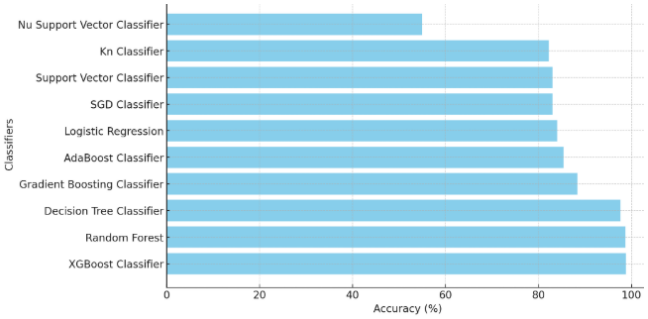


Fig. 2. Accuracy of employee attrition prediction models on dataset [31].

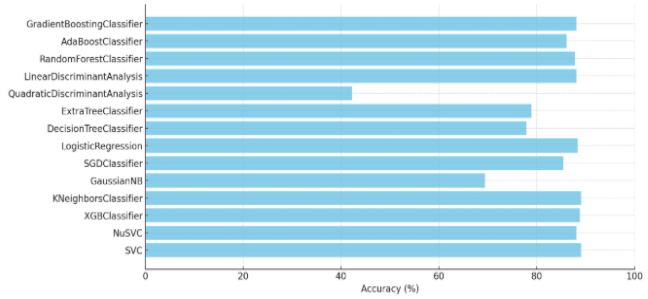


Fig. 3. Accuracy of employee attrition prediction models on dataset [47].

The dataset used for employee attrition prediction in [39]. contains 35 features, including attrition, age, business travel, distances from home, daily rates, education, and more. These features represent various aspects of employees' demographics, job roles, and satisfaction levels, which can influence their decision to stay or leave. Key attributes like job happiness and balance between work and personal life were identified as the most significant predictors of attrition in this dataset, aligning with broader HR analytics trends that emphasize employee well-being as a retention driver [4], [48]. Nawaz et al. (2024) [25] underscore that factors such as job satisfaction and work-life balance are consistently the top predictors of attrition across various industries, highlighting the importance of focusing on employee well-being in retention strategies [25], [48]. The accuracy achieved by the model is approximately 85.37%, which is a relatively good result, though further model optimization could improve this. Identifying the most important features helps organizations focus on retention strategies effectively.

Choosing the right model for employee turnover prediction depends on accuracy, computational efficiency, and interpretability. Random Forest and XGBoost achieve the highest accuracy but are computationally demanding. For faster execution at lower cost, Logistic Regression and SVC are practical alternatives. If transparency is a priority, Decision Tree is the best choice. On the other hand, NuSVC and Gaussian Naïve Bayes have poor performance and are not

Authorized licensed use limited to: University of Thrace (Democritus University of Thrace). Downloaded on November 07,2025 at 19:20:19 UTC from IEEE Xplore. Restrictions apply.

recommended. The final choice depends on the needs of the business.

VII. Improvements and Future Prospects

A key issue that needs improvement concerns the generalizability of the models. The development of adaptive models could improve prediction accuracy in sectors such as healthcare, education, and technology. Furthermore, most models are based exclusively on structured historical data, ignoring information from unstructured sources such as employee feedback and internal communications. The integration of NLP and sentiment analysis can allow for the early detection of employee dissatisfaction. Another critical point is the interpretability of the models, as HR managers find it difficult to understand the decisions of the systems. Counterfactuals and causal inference can improve the transparency and acceptance of these models by organizations. At the same time, ethical and privacy issues remain critical, as employee turnover prediction requires the processing of sensitive information. The use of decentralized techniques, such as Federated Learning, can ensure compliance with privacy regulations without reducing the effectiveness of predictions. Finally, future research should focus on developing targeted interventions for human resource retention, through personalized strategies.

VIII. Conclusions

The integration of machine learning into employee retention strategies represents a paradigm shift, enabling organizations to anticipate integration of machine learning into employee retention strategies represents a paradigm shift, enabling organizations to anticipate and address attrition challenges proactively. Advanced models, like Random Forest and Gradient Boosted Tree, deliver accurate predictions by uncovering complex data patterns, while deep learning methods like LSTM networks offer unique advantages in analyzing sequential data. Explainable AI tools further empower HR teams to interpret model outputs transparently, aligning decisions with organizational and ethical standards.

Moreover, multimodal approaches that incorporate qualitative and quantitative data provide a comprehensive understanding of employee behavior. This holistic perspective allows for the design of more targeted and effective retention strategies. By adopting these innovative technologies, organizations can move from reactive to predictive management of workforce attrition, resulting in increased employee satisfaction and lower turnover costs. Moreover, the use of multimodal data enriches the understanding of attritional factors, facilitating tailored, effective interventions.

Future research should also explore integrating real-time analytics powered by edge computing to enhance the timeliness of retention strategies. This would enable organizations to respond dynamically to emerging trends and changes in employee sentiment. Future research should also prioritize extending the application of ML tools to small and medium-sized enterprises, integrating explainability frameworks for broader usability, and addressing data privacy concerns through advanced security methods. These efforts will help organizations optimize their workforce management practices and adapt to evolving challenges in the labor market.

Acknowledgment

This work was supported by the MPhil program “Advanced Technologies in Informatics and Computers”, which was hosted by the Department of Informatics, Democritus University of Thrace, Kavala, Greece.

References

[1] A. Raza, K. Munir, M. Almutairi, F. Younas, and M. M. S. Fareed, “Predicting Employee Attrition Using Machine Learning Approaches,” *Appl. Sci.*, vol. 12, no. 13, p. 6424, Jun. 2022, doi: 10.3390/app12136424.

[2] A. I. Al-Alawi and Y. A. Ghanem, “Predicting Employee Attrition Using Machine Learning: A Systematic Literature Review,” in *2024 ASU International Conference in Emerging Technologies for Sustainability and Intelligent Systems (ICETSIS)*, IEEE, Jan. 2024, pp. 526–530. doi: 10.1109/ICETSIS61505.2024.10459451.

[3] R. Jain and A. Nayyar, “Predicting Employee Attrition using XGBoost Machine Learning Approach,” in *2018 International Conference on System Modeling & Advancement in Research Trends (SMART)*, IEEE, Nov. 2018, pp. 113–120. doi: 10.1109/SYSMART.2018.8746940.

[4] A. Chaurasia, S. Kadam, K. Bhagat, S. Gauda, and P. Shingane, “Employee Attrition Prediction using Artificial Neural Networks,” in *2023 4th International Conference for Emerging Technology (INCET)*, IEEE, May 2023, pp. 1–6. doi: 10.1109/INCET57972.2023.10170676.

[5] M. Al Akasheh, E. F. Malik, O. Hujran, and N. Zaki, “A decade of research on machine learning techniques for predicting employee turnover: A systematic literature review,” *Expert Syst. Appl.*, vol. 238, p. 121794, Mar. 2024, doi: 10.1016/j.eswa.2023.121794.

[6] C. Zhang and W. Han, “Ensembles of decision trees and gradient-based learning for employee turnover rate prediction,” *PeerJ Comput. Sci.*, vol. 10, p. e2387, Oct. 2024, doi: 10.7717/peerj-cs.2387.

[7] S. Barman *et al.*, “A Two-Stage Stacking Ensemble Learning for Employee Attrition Prediction,” 2025, pp. 119–132. doi: 10.1007/978-981-97-3937-0_9.

[8] R. Caruana and A. Niculescu-Mizil, “An empirical comparison of supervised learning algorithms,” in *Proceedings of the 23rd international conference on Machine learning - ICML ’06*, New York, New York, USA: ACM Press, 2006, pp. 161–168. doi: 10.1145/1143844.1143865.

[9] S. Polisetti, M. Bhargavi, S. Chitneni, S. Eluri, N. Kattamuri, and R. R, “Stacking Models for Employee Attrition Prediction: Leveraging Logistic Regression and Random Forest,” in *2024 8th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)*, IEEE, Oct. 2024, pp. 863–867. doi: 10.1109/I-SMAC61858.2024.10714670.

[10] L. Sun, J. Du, L.-R. Dai, and C.-H. Lee, “Multiple-target deep learning for LSTM-RNN based speech enhancement,” in *2017 Hands-free Speech Communications and Microphone Arrays (HSCMA)*, IEEE, 2017, pp. 136–140. doi: 10.1109/HSCMA.2017.7895577.

[11] G. Sharma, “A literature review on application of Artificial Intelligence in Human Resource Management and its practices in current organizational scenario,” in *2021 Fifth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)*, IEEE, Nov. 2021, pp. 594–600. doi: 10.1109/I-SMAC52330.2021.9640954.

[12] S. R. Seelam, K. H. Kumar, M. S. Supritha, G. Gnaneswar, and V. V. M. Reddy, “Comparative Study of Predictive Models to Estimate Employee Attrition,” in *2022 7th International Conference on Communication and Electronics Systems (ICCES)*, IEEE, Jun. 2022, pp. 1602–1607. doi: 10.1109/ICCES54183.2022.9835964.

[13] C. F. Olson, “Parallel algorithms for hierarchical clustering,” *Parallel Comput.*, vol. 21, no. 8, pp. 1313–1325, Aug. 1995, doi: 10.1016/0167-8191(95)00017-I.

[14] P. Kumar, S. B. Gaikwad, S. T. Ramya, T. Tiwari, M. Tiwari, and B. Kumar, “Predicting Employee Turnover: A Systematic Machine Learning Approach for Resource Conservation and Workforce Stability,” in *RAiSE-2023*, Basel Switzerland: MDPI, Dec. 2023, p. 117. doi: 10.3390/engproc2023059117.

[15] S. N. Khera and Divya, “Predictive Modelling of Employee Turnover in Indian IT Industry Using Machine Learning Techniques,” *Vis. J. Bus. Perspect.*, vol. 23, no. 1, pp. 12–21, Mar. 2019, doi: 10.1177/0972262918821221.

Authorized licensed use limited to: University of Thrace (Democritus University of Thrace). Downloaded on November 07,2025 at 19:20:19 UTC from IEEE Xplore. Restrictions apply.

[16] K. Ravesangar and S. Narayanan, "Adoption of HR analytics to enhance employee retention in the workplace: A review," *Hum. Resour. Manag. Serv.*, vol. 6, no. 3, p. 3481, Aug. 2024, doi: 10.18282/hrms.v6i3.3481.

[17] L. Gu, X. Zhang, S. You, S. Zhao, Z. Liu, and T. Harada, "Semi-Supervised Learning in Medical Images Through Graph-Embedded Random Forest," *Front. Neuroinform.*, vol. 14, Nov. 2020, doi: 10.3389/fninf.2020.601829.

[18] Z. Taner, O. Areta Hızıroğlu, and K. Hızıroğlu, "Leveraging Machine Learning Methods for Predicting Employee Turnover Within the Framework of Human Resources Analytics," *J. Intell. Syst. Theory Appl.*, vol. 7, no. 2, pp. 145–158, Sep. 2024, doi: 10.38016/jista.1440879.

[19] G. Marín Díaz, J. J. Galán Hernández, and J. L. Galdón Salvador, "Analyzing Employee Attrition Using Explainable AI for Strategic HR Decision-Making," *Mathematics*, vol. 11, no. 22, p. 4677, Nov. 2023, doi: 10.3390/math11224677.

[20] A. Abonamah, D. La Torre, M. Poulin, and M. Repetto, "Explainable Artificial Intelligence in Human Resources: a Computational Study," in *2022 International Conference on Data Analytics for Business and Industry (ICDABI)*, IEEE, Oct. 2022, pp. 1–6. doi: 10.1109/ICDABI56818.2022.10041624.

[21] E. Knudsen, "Supervised learning in the brain," *J. Neurosci.*, vol. 14, no. 7, pp. 3985–3997, Jul. 1994, doi: 10.1523/JNEUROSCI.14-07-03985.1994.

[22] L. Geiler, S. Affeldt, and M. Nadif, "A survey on machine learning methods for churn prediction," *Int. J. Data Sci. Anal.*, vol. 14, no. 3, pp. 217–242, Sep. 2022, doi: 10.1007/s41060-022-00312-5.

[23] X. Wang, X. Lin, and X. Dang, "Supervised learning in spiking neural networks: A review of algorithms and evaluations," *Neural Networks*, vol. 125, pp. 258–280, May 2020, doi: 10.1016/j.neunet.2020.02.011.

[24] F. Fallucchi, M. Coladangelo, R. Giuliano, and E. William De Luca, "Predicting Employee Attrition Using Machine Learning Techniques," *Computers*, vol. 9, no. 4, p. 86, Nov. 2020, doi: 10.3390/computers9040086.

[25] M. S. Nawaz, M. Z. Nawaz, P. Fournier-Viger, and J. M. Luna, "Analysis and classification of employee attrition and absenteeism in industry: A sequential pattern mining-based methodology," *Comput. Ind.*, vol. 159–160, p. 104106, Aug. 2024, doi: 10.1016/j.compind.2024.104106.

[26] V. K. Kambhampati and K. B. V. B. Rao, "Advancing Employee Attrition Models: A Systematic Review of Machine Learning Techniques and Emerging Research Opportunities," in *2024 8th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)*, IEEE, Oct. 2024, pp. 1100–1107. doi: 10.1109/I-SMAC61858.2024.10714595.

[27] S. Na, L. Xumin, and G. Yong, "Research on k-means Clustering Algorithm: An Improved k-means Clustering Algorithm," in *2010 Third International Symposium on Intelligent Information Technology and Security Informatics*, IEEE, Apr. 2010, pp. 63–67. doi: 10.1109/IITSI.2010.74.

[28] I. Ullah, B. Raza, A. K. Malik, M. Imran, S. U. Islam, and S. W. Kim, "A Churn Prediction Model Using Random Forest: Analysis of Machine Learning Techniques for Churn Prediction and Factor Identification in Telecom Sector," *IEEE Access*, vol. 7, pp. 60134–60149, 2019, doi: 10.1109/ACCESS.2019.2914999.

[29] T. Vafeiadis, K. I. Diamantaras, G. Sarigiannidis, and K. C. Chatzisavvas, "A comparison of machine learning techniques for customer churn prediction," *Simul. Model. Pract. Theory*, vol. 55, pp. 1–9, Jun. 2015, doi: 10.1016/j.simpat.2015.03.003.

[30] S. Selvin, R. Vinayakumar, E. A. Gopalakrishnan, V. K. Menon, and K. P. Soman, "Stock price prediction using LSTM, RNN and CNN-sliding window model," in *2017 International Conference on Advances in Computing, Communications and Informatics (ICACCI)*, IEEE, Sep. 2017, pp. 1643–1647. doi: 10.1109/ICACCI.2017.8126078.

[31] O. Iparraguirre-Villanueva, L. Chauca-Huete, R. Prieto-Chavez, and C. Paulino-Moreno, "Employee attrition prediction using machine learning models," in *Proceedings of the 22nd LACCEI International Multi-Conference for Engineering, Education and Technology (LACCEI 2024): "Sustainable Engineering for a Diverse, Equitable, and Inclusive Future at the Service of Education, Research, and Industry for a Societ*, Latin American and Caribbean Consortium of Engineering Institutions, 2024. doi: 10.18687/LACCEI2024.1.1.498.

[32] S. Krishna and S. Sidharth, "Analyzing Employee Attrition Using Machine Learning: the New AI Approach," in *2022 IEEE 7th International conference for Convergence in Technology (I2CT)*, IEEE, Apr. 2022, pp. 1–14. doi: 10.1109/I2CT54291.2022.9825342.

[33] M. Gollapalli, A.-U. Rahman, A. Osama, A. Alfaify, M. Yassin, and A. Alabdullah, "Data Mining and Visualization to Understand Employee Attrition and Work Performance," in *2023 3rd International Conference on Computing and Information Technology (ICCIT)*, IEEE, Sep. 2023, pp. 149–154. doi: 10.1109/ICCIT58132.2023.10273889.

[34] E. F. R. Seixas, J. Viterbo, F. Bernardini, F. Seixas, and C. Pantoja, "Applying Artificial Intelligence for Talent Retention: a Systematic Literature Review," in *2023 18th Iberian Conference on Information Systems and Technologies (CISTI)*, IEEE, Jun. 2023, pp. 1–6. doi: 10.23919/CISTI58278.2023.10211299.

[35] P. Tavakoli-Kolour *et al.*, "Genetic species identification and population structure of grouper Epinephelus coioides (Hamilton, 1822) collected from fish markets along the Persian Gulf and the Oman Sea," *PeerJ*, vol. 10, p. e14179, Oct. 2022, doi: 10.7717/peerj.14179.

[36] S. De Vos, C. Bockel-Rickermann, J. Van Belle, and W. Verbeke, "Predicting Employee Turnover: Scoping and Benchmarking the State-of-the-Art," *Bus. Inf. Syst. Eng.*, Oct. 2024, doi: 10.1007/s12599-024-00898-z.

[37] M. R. Perello Marin and M. Tuffaha, "Artificial intelligence definition, applications and adoption in Human Resource Management: a systematic literature review," *Int. J. Bus. Innov. Res.*, vol. 1, no. 1, p. 1, 2021, doi: 10.1504/IJBIR.2021.10040005.

[38] V. Nasteski, "An overview of the supervised machine learning methods," *HORIZONS.B*, vol. 4, pp. 51–62, Dec. 2017, doi: 10.20544/HORIZONS.B.04.1.17.P05.

[39] The clever programmer, "Employee Attrition Prediction with Python." Accessed: Jan. 05, 2025. [Online]. Available: https://thecleverprogrammer.com/2020/11/21/employee-attrition-prediction-with-python/

[40] N. Darapaneni *et al.*, "A Detailed Analysis of AI Models for Predicting Employee Attrition Risk," in *2022 IEEE 10th Region 10 Humanitarian Technology Conference (R10-HTC)*, IEEE, Sep. 2022, pp. 243–246. doi: 10.1109/R10-HTC54060.2022.9929893.

[41] S. Al-Darraji, D. G. Honi, F. Fallucchi, A. I. Abdulsada, R. Giuliano, and H. A. Abdulmalik, "Employee Attrition Prediction Using Deep Neural Networks," *Computers*, vol. 10, no. 11, p. 141, Nov. 2021, doi: 10.3390/computers10110141.

[42] Nancy, S. Sil, R. Rani, N. Bisla, G. Jaiswal, and A. Sharma, "Unleashing the Power of Machine Learning: Advancing Early Prediction and Analysis of Diabetes Mellitus," in *2023 World Conference on Communication & Computing (WCONF)*, IEEE, Jul. 2023, pp. 1–5. doi: 10.1109/WCONF58270.2023.10235242.

[43] Y. Xie, Z. Xu, J. Zhang, Z. Wang, and S. Ji, "Self-Supervised Learning of Graph Neural Networks: A Unified Review," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2412–2429, Feb. 2023, doi: 10.1109/TPAMI.2022.3170559.

[44] S. Pereira, J. de Lima, A. Vieira, and W. Brandão, "Exploiting Data Spatial Dependencies for Employee Turnover Prediction," in *Proceedings of the 20th International Conference on Web Information Systems and Technologies*, SCITEPRESS - Science and Technology Publications, 2024, pp. 250–257. doi: 10.5220/0012948600003825.

[45] M. R. Shafie, H. Khosravi, S. Farhadpour, S. Das, and I. Ahmed, "Enhancing Employee Turnover Prediction with Optimized Anns and Data Augmentation: A Cluster-Based Approach in Hr Analytics." 2024. doi: 10.2139/ssrn.4698645.

[46] H. U. Dike, Y. Zhou, K. K. Deveerasetty, and Q. Wu, "Unsupervised Learning Based On Artificial Neural Network: A Review," in *2018 IEEE International Conference on Cyborg and Bionic Systems (CBS)*, IEEE, Oct. 2018, pp. 322–327. doi: 10.1109/CBS.2018.8612259.

[47] V. Adep, "Employee Attrition Prediction using ML: IBM_HR_Analytics_Employee_Attrition_Performance," Kaggle. Accessed: Jan. 15, 2025. [Online]. Available: https://www.kaggle.com/code/adepvenugopal/employee-attrition-prediction-using-ml

[48] P. K. Jain, M. Jain, and R. Pamula, "Explaining and predicting employees' attrition: a machine learning approach," *SN Appl. Sci.*, vol. 2, no. 4, p. 757, Apr. 2020, doi: 10.1007/s42452-020-2519-4.

Authorized licensed use limited to: University of Thrace (Democritus University of Thrace). Downloaded on November 07,2025 at 19:20:19 UTC from IEEE Xplore. Restrictions apply.